Quality Function Deployment Chart

Project title: BME 362 QFD

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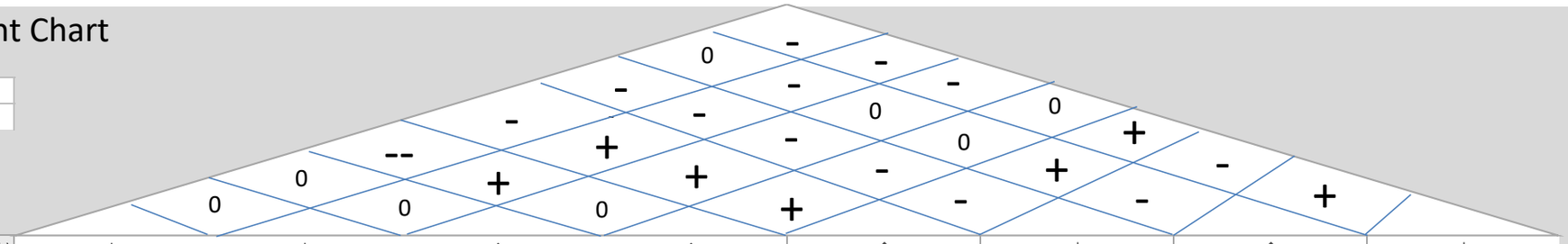

	Customer importance rating	Functional Requirements (How's) → / Customer Requirements - (What's) ↓	Minimize wearability burden	Minimize system response latency	Maximize automation of incident logging	Maximize device operational reliability	Maximize fall detection performance	Minimize total device cost	Maximize fall type coverage	Minimize false alarm rate	Apple Watch	Alert Pendant
		Desired direction of improvement (↑,0,↓)	↓	↓	↑	↑	↑	↓	↑	↓		
1	5	Comfortable to wear continuously	9	0	0	0	0	3	0	0	2	2
2	4	Minimal device maintenance and repair	1	0	0	9	0	3	0	0	4	2
3	2	Reduces documentation burden	0	0	9	0	0	0	1	0	1	1
4	5	Detect multiple fall types	0	0	3	0	3	0	9	3	0	0
5	5	Reduces lie time measurably	0	9	0	0	9	1	0	3	2	2
	5	Alerts arrive quickly	0	9	0	1	9	0	0	0	1	2
7	4	Reduces alarm fatigue	0	0	0	0	1	0	0	9	1	0
8	4	Device is easily accessible/affordable	3	1	0	3	3	9	1	1	0	4
		Technical importance score	61	94	33	53	121	68	51	70	SUM: 551	
		Importance %	11%	17%	6%	10%	22%	12%	9%	13%		
		Priorities rank	**5**	**2**	**8**	**6**	**1**	**4**	**7**	**3**		
		Marginal Threshold	75	60	30	1	90	60	2	10		
		Ideal Value	50	30	25	2	100	50	4	0		
		Unit of Measurement	g	s	min (mean time needed to complete an incident report)	m (max height it can be dropped from without breaking)	% (unitless true positive rate)	CAD ($)	unitless (number of unique fall types)	% (unitless false positive rate)		

VirtuSense VSTAlert
5
1
4
5
3
5
4
0